Journal of Molecular Spectroscopy 344 (2018) 17–20

Contents lists available at ScienceDirect

Journal of Molecular Spectroscopy

journal homepage: www.elsevier.com/locate/jms

Note

Spectroscopy of the low-lying states of CaO^+

Robert A. VanGundy, Joshua H. Bartlett, Michael C. Heaven *

Department of Chemistry, Emory University, Atlanta, GA 30322, United States

ARTICLE INFO

Article history:
Received 16 August 2017
In revised form 29 September 2017
Accepted 2 October 2017
Available online 3 October 2017

ABSTRACT

Low energy states of CaO^+ have been examined by means of two-color threshold ionization spectroscopy. It is established that the ground state is $X^2\Pi_i$ and the first excited state, just 685 cm^{-1} higher in energy, is $A^2\Sigma^+$. Due to the close proximity of the $X^2\Pi_{1/2}$ and $A^2\Sigma^+$ vibronic levels, they are mutually perturbing via the spin-orbit operator. A deperturbation analysis is presented for two pairs of levels. These measurements also yielded improved values for the ionization energy of CaO ($55,428 \pm 3$ cm^{-1}) and the bond dissociation energy of CaO^+ ($27,000 \pm 600$ cm^{-1}). The results validate previous theoretical calculations and provide spectroscopic data of value for future ion trap studies of CaO^+ at ultra-cold temperatures.

1. Introduction

The ionization energy (IE) of CaO and the electronic structure of the CaO^+ cation are topics that have attracted the attention of both experimentalists and theoreticians [1–6]. The IE of CaO has been measured using electron impact techniques, which yielded values in the 6.5–6.9 eV range [1,5,6]. High-level electronic structure calculations have given IE values that were in respectable agreement [2–4]. Apart from the detection by mass spectrometry, there are no published experimental data for the CaO^+ ion.

Systematic theoretical calculations have been carried out for the family of alkaline earth oxide ions MO^+ with M = Be, Mg, Ca, Sr, and Ba [2]. With the exception of BeO^+, the bonding in these molecules is markedly ionic. Formally they have the configuration $M^{2+}O(2p^5)^-$. The lowest energy states are $^2\Sigma^+$ and $^2\Pi$, corresponding to the half-filled O 2p orbital being oriented towards the M^{2+} ion or perpendicular to the bond. For the lighter metals the ground state is predicted to be $X^2\Pi$, but this changes to $X^2\Sigma^+$ for Sr and Ba. Partridge et al. [2] attributed this switch to a competition between the electrostatic attraction and the more rapidly increasing Pauli repulsion. In a recent spectroscopic study of BaO^+ [7] we confirmed that the ground state was $X^2\Sigma^+$ and observed the origin of the $A^2\Pi_{3/2}$ state at 1482 cm^{-1}. Calculations for CaO^+ predict a $X^2\Pi$ ground state with the $A^2\Sigma^+$ state at approximately 650 cm^{-1} [2–4]. The most detailed calculations have been reported by Khalil et al. [4] They applied the multi-reference configuration interaction method with the Davidson correction (MRCI + Q) to CaO^-, CaO and CaO^+. The basis sets were of 5-zeta quality (cc-pCV5Z). As the $2p^5$ sub-shell of O^- is more than half-filled the spin-orbit splitting of $X^2\Pi$ is inverted, with the $|\Omega| = 3/2$ component below $|\Omega| = 1/2$ (where Ω is the projection of the electronic angular momentum along the bond axis). For CaO^+, the spin-orbit splitting of the $^2\Pi$ state, and the spin-orbit induced perturbations between $X^2\Pi_{1/2}$ and $A^2\Sigma^+$ were considered. Khalil et al. [4] predicted an energy interval of 654.9 cm^{-1} between $X^2\Pi_{3/2}(v'' = 0)$ and $A^2\Sigma^+(v' = 0)$.

In addition to the elucidation of electronic structure, the CaO^+ and BaO^+ ions are of interest for studies of molecular ions at ultra-cold temperatures. The laser cooling and trapping techniques that have been successfully applied to the alkali atoms are also applicable to the isoelectronic alkaline earth ions [8–12]. An advantage of the latter is that they can be held and manipulated by externally applied fields. Hence, Coulomb crystals of Ca^+ and Ba^+ ions are readily formed, and the individual ions within the trap can be imaged using the photons emitted from the laser cooling process. Controlled reactions can be induced by adding trace amounts of oxidants that react to form simple molecular ions (e.g., $Ca^+ + O_2 \rightarrow CaO^+ + O$) [11,12]. These species no longer absorb the cooling laser light, so the presence of the molecular ion is indicated by the positions and frequencies of the atomic ions that are adjacent to the dark spot. Ejection of the ions from the trap into a mass spectrometer confirms the identity of the molecular ions [10].

In these experiments, it is evident that the translation motion of the molecular ions is cooled by the interactions with surrounding laser-cooled atomic ions [10]. However, the internal energy content of the ions is not known. Due to the small numbers of ions used in these experiments, spectroscopic probes of the internal state distributions require highly sensitive detection techniques. Resonantly enhanced multi-photon dissociation (REMPD) combined with mass spectrometric detection of the resulting atomic ions can provide the required sensitivity [9,10,13,14]. One of the challenges for any spectroscopic detection techniques has been

* Corresponding author.
E-mail address: mheaven@emory.edu (M.C. Heaven).

https://doi.org/10.1016/j.jms.2017.10.001

the lack of spectroscopic data for the relevant molecular ions. Our recent studies of the BaO^+ [7] and $BaCl^+$ [15] ions were, in part, motivated by the need for these data. When spectroscopic data are not available, experimentalists have relied on electronic structure calculations to guide the search for suitable vibronic transitions. Consequently, evaluating computationally determined molecular constants against experimental data helps to determine the sophistication level of theoretical calculations required to obtain suitably reliable predictions.

In the present study we report the first experimentally determined spectroscopic constants for the $X^2\Pi$ and $A^2\Sigma^+$ states of CaO^+, and an accurate IE for CaO. The results are discussed in the context of previous experimental and theoretical studies.

2. Experimental

The apparatus used for these measurements has been described previously [16]. CaO was produced in the gas phase by pulsed laser vaporization of Ca in the presence of a He/O_2 gas mixture (0.1% O_2). The gas mixture was subject to supersonic expansion, in order to cool the reaction products. Resonantly enhanced two-color photoionization was used to examine the low energy states of CaO^+. This was accomplished using two independently tunable pulsed dye lasers (with linewidths of 0.15 and 0.3 cm^{-1}, FWHM). The first laser was tuned to one of five different vibronic bands of CaO in the 31,970–33,680 cm^{-1} range. All but one of these bands had been characterized previously in preparation for these measurements [17]. Laser induced fluorescence (LIF) spectroscopy was used to optimize the tuning of the first laser. Once a specific resonance had been established, the second laser was tuned over the wavelength range where the sum of the photon energies would span the ionization threshold. Previous studies indicated an IE in the range 52,400–55,700 cm^{-1}. A time-of-flight mass spectrometer was used to monitor the flux of CaO^+ ions, thereby producing a photoionization efficiency (PIE) spectrum. The IE was further refined by recording the pulsed field ionization – zero kinetic energy (PFI-ZEKE) photoelectron spectrum. For these measurements the dye lasers were used to excite CaO to long-lived, high-n Rydberg states that lie just below the ionization threshold for each state of the cation. This was done under field-free conditions. After a delay of 2 μs a pulsed field of 0.54 V/cm was applied to ionize the excited CaO, and convey the electrons to a microchannel plate detector. The absolute wavenumber calibration of the second dye laser was established against a commercial wavemeter (Bristol Instruments model 821) and by recording two-photon transitions of atomic Ca via one-color, 2 + 1 photo-ionization.

3. Results

A representative LIF spectrum for jet-cooled CaO is shown in Fig. 1. This is the $C^1\Sigma^+$ - $X^1\Sigma^+$ 6-0 band, with a rotational line intensity contour that was consistent with a temperature of 10 K. Survey scans of the LIF spectrum indicated that a minor fraction of the population was in $v''>0$ levels. All of the PFI-ZEKE measurements reported here involved excitation sequences that originated from $X^1\Sigma^+$ $v''=0$. Note that the most intense feature of Fig. 1 is the R-branch band head, which occurred just 1.7 cm^{-1} above the band origin. This was typical for CaO bands in the 30,000–34,000 cm^{-1} range [17], and all of the two photon excitation measurements were performed with the first laser tuned to the R-branch band head of the chosen vibronic band.

A low-resolution determination of the IE for CaO was made using initial excitation of the $F^1\Pi$-$X^1\Sigma^+$ 12-0 band ($T_0 = 32646.0$ cm^{-1}). PIE scans, with mass selected detection of the CaO^+ ion, showed a clear onset of signal at an energy of 55,436(30) cm^{-1}

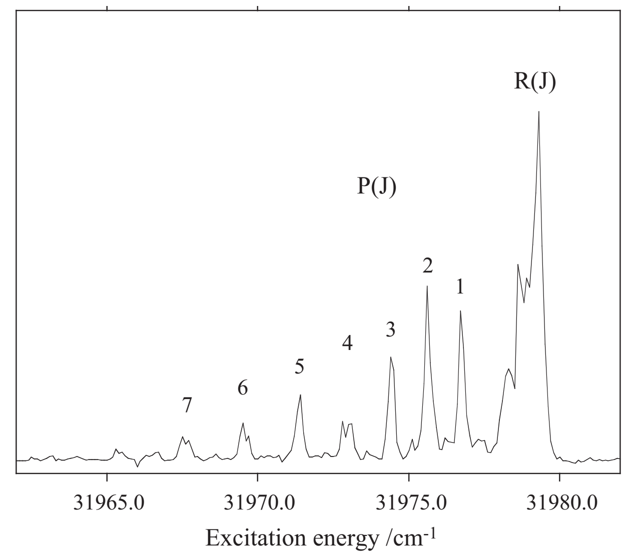

Fig. 1. Laser induced fluorescence spectrum of the CaO $C^1\Sigma^+$-$X^1\Sigma^+$ 6-0 band.

(where the 1-σ error in the least significant digits is given in parentheses). This is a field-free value where the energy lowering caused by the electric field of the mass spectrometer has been taken into account.

PFI-ZEKE spectra for CaO^+ were recorded using the excitation schemes given in Table 1. Fig. 2 shows a survey scan covering the energy range from the IE to levels lying 2010 cm^{-1} above the ground state zero-point energy of CaO^+. The band center energies are listed in Table 2. These values have been field-corrected corrected by adding 4.4 cm^{-1} to the sums of the two-photon energies. Based on the previous theoretical studies [2–4], the structure of this spectrum is readily assigned. The first excited level is 130.5 cm^{-1} above the origin band, so this is clearly the upper spin-orbit component of the $X^2\Pi$ state. Similar pairs of transitions were observed at higher energies, with energy spacings (relative to the origin) that were quite close to those predicted by Khalil et al. [4] (which are given in parentheses in Table 2). The first level of the $A^2\Sigma^+$ state was found at 685.2 cm^{-1}. An expanded scan of the origin band is shown in Fig. 3. The rotational lines seen in this trace can be assigned to the $J = 5.5$–8.5 levels, indicating that the first photon was primarily exciting the R(5) rotational line of the C-X 6-0 band. Fitting to the rotational lines yielded a rotational constant for $X^2\Pi_{3/2}$ $v''=0$ of $B_0 = 0.368(9)$ cm^{-1}. This was consistent with the predicted value for B_e of 0.3725 cm^{-1}[4]. Assuming that the most intense line in this trace corresponds to the $\Delta N = 0$ transition (where N is the angular momentum exclusive of electron spin), the IE of CaO is found to be 55,428(3) cm^{-1}.

Khalil et al. [4] noted that the perturbations between the $X^2\Pi_{1/2}$ and $A^2\Sigma^+$ vibronic levels caused shifts that were as large as 6 cm^{-1}, while the vibrational levels of the $X^2\Pi_{3/2}$ state were unperturbed.

Table 1
Excitation sequences used to observe the low energy states of CaO^+.

Electronic state[a]	v′	Band origin[b]	Ion states accessed
$C^1\Sigma^+$	6	31977.6	$X^2\Pi$, v = 0
$C^1\Sigma^+$	7	32513.5	$X^2\Pi$, v = 1; $A^2\Sigma^+$, v = 0
$F^1\Pi$	13	33101.2	$X^2\Pi$, v = 2; $A^2\Sigma^+$, v = 1
$U^1\Sigma^+$	u	33670.8	$X^2\Pi$, v = 3

[a] All transitions originated from the $X^1\Sigma^+$, v = 0 level. U is a previously unidentified state. The vibrational quantum number for this band has not been determined.
[b] Band origins for the first photon transitions, given in cm^{-1} units. The origins for the C and F states are from Ref. [17].

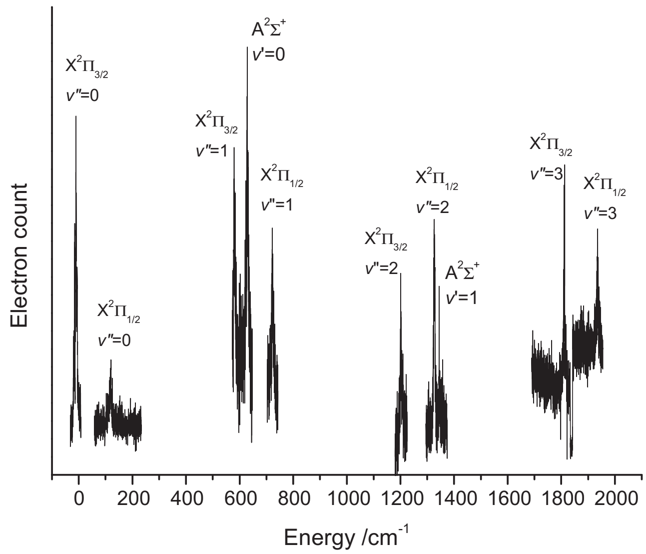

Fig. 2. PFI-ZEKE survey scan showing the low energy vibronic states of CaO^+.

Table 2
Observed and calculated energies of the low-lying states of CaO^+.

v	$X^2\Pi_{3/2}$	$X^2\Pi_{1/2}$	$A^2\Sigma^+$
0	0	130.5	685.2
	(0)	(125.2)	(654.9)
1	636.0	778.3	1406.1
	(633.8)	(766.8)	(1358.0)
2	1263.4	1388.4	–
	(1262.1)	(1390.2)	(2053.3)
3	1878.6	2002.3	–
	(1885.0)	(2009.9)	(2738.8)

Energies are given in cm^{-1} units. The calculated energies, taken from Ref. [4], are given in parentheses.

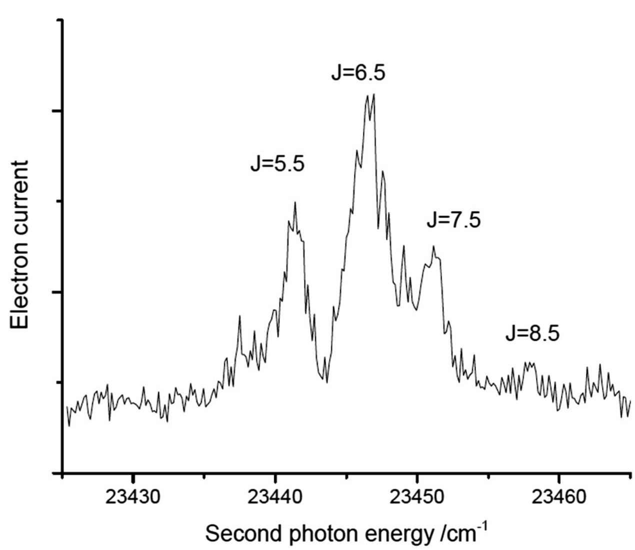

Fig. 3. Expanded view of the PFI-ZEKE spectrum for the $X^2\Pi_{3/2}$, $v''=0$ level of CaO^+. The first laser was tuned to the R-branch head of the $C^1\Sigma^+$-$X^1\Sigma^+$ 6-0 band.

The vibronic state energies presented in Table 2 were qualitatively consistent with this behavior. The vibrational energies of the $X^2\Pi_{3/2}$ state ($G(v)$) were well represented by the Morse energy level expression $G(v) = \omega_e v - \omega_e x_e v(v+1)$, with $\omega_e = 646.9(9)$ and $\omega_e x_e = 5.2(3)$ cm^{-1}.

To unravel the perturbations between the $X^2\Pi_{1/2}$ and $A^2\Sigma^+$ vibronic levels we began by assuming that the perturbation of the $X^2\Pi_{1/2}$, $v''=0$ was not significant at the resolution of our measurements, and that the spin-orbit coupling constant would be nearly independent of the vibrational energy for $v''=0$–3. The latter assumption was supported by the calculations of Khalil et al. [4], where the spin-orbit interval between the unperturbed levels of the $X^2\Pi$ state varied from 125.4 to 125.5 cm^{-1} for $v''=0$–3. Hence, the deperturbed energies of the $X^2\Pi_{1/2}$ state were estimated by adding the measured spin-orbit interval of 130.5 cm^{-1} to the observed energy levels of $X^2\Pi_{3/2}$. Our estimates for the deperturbed energies are given in Table 3. Next, we made the assumption that the perturbations were dominated by two-state interactions (the close-lying $X^2\Pi_{1/2}$ $v''=1/A^2\Sigma^+$ $v'=0$ and $X^2\Pi_{1/2}$ $v''=2/A^2\Sigma^+$ $v'=1$ pairs). Deperturbed energies for the $A^2\Sigma^+$ levels were then determined by applying the energy shifts determined from the $X^2\Pi_{1/2}$ manifold. For example, mixing with $A^2\Sigma^+$, $v'=0$ pushes the $X^2\Pi_{1/2}$, $v''=1$ level up by 11.8 cm^{-1}. Hence it was estimated that the observed $A^2\Sigma^+$, $v=0$ level had been pushed down 11.8 cm^{-1} below the unperturbed energy. The shift experienced by the $X^2\Pi_{1/2}$ $v=2/A^2\Sigma^+$ $v=1$ pair was similarly estimated to be ± 5.5 cm^{-1} (see Table 3).

The A-X perturbation matrix elements were determined by solving the two-state perturbation model

$$\begin{pmatrix} E(X,v'') \\ E(A,v\prime) \end{pmatrix} = \begin{pmatrix} E^0(X,v'') & H_{A,X}(v',v'') \\ H_{X,A}(v'',v\prime) & E^0(A,v\prime) \end{pmatrix} \tag{1}$$

where $E^0(X, v'')$ and are $E^0(A, v')$ are the deperturbed energies, $E(X, v'')$ and $E(A, v')$ are the observed energies, and the $H_{A,X}(v',v'')/H_{X,A}(v'',v')$ terms are the perturbation matrix elements. We assume that the latter are real, so that $H_{A,X}(v',v'') = H_{X,A}(v'',v\prime)$. Analysis of the interactions between the $X^2\Pi_{1/2}$ $v=1/A^2\Sigma^+$ $v'=0$ and $X^2\Pi_{1/2}$ $v=2/A^2\Sigma^+$ $v''=1$ pairs defined perturbation matrix elements of $|H_{A,X}(v',v'')|$ of 31.0 and 8.2 cm^{-1}, respectively.

The off-diagonal matrix elements of Eq. (1) are defined by the product

$$H_{A,X}(v',v'') = \langle A|H_{SO}|X\rangle\sqrt{F_{v',v''}} \tag{2}$$

where $\langle A|H_{SO}|X\rangle$ is the electronic matrix element of the spin-orbit operator and $F_{v',v''}$ is the Franck-Condon factor (FCF). To obtain the electronic matrix element from this relationship we have calculated the A-X FCF's using the program Level 8.0 [18]. The potential energy curves were approximated by Morse functions where the parameters were determined from the calculated molecular constants of Khalil et al. [4] The resulting FCF's and the potential energy curve parameters are given in Table 4. Application of Eq. (2) then yielded $|\langle A|H_{SO}|X\rangle|$ values of 57.4 and 64.8 cm^{-1} for the $X^2\Pi_{1/2}$ $v=1/A^2\Sigma^+$ $v'=0$ and $X^2\Pi_{1/2}$ $v=2/A^2\Sigma^+$ $v''=1$ pairs. The latter value is very sensitive to the small FCF for the $v'=2/v''=1$ overlap. Increasing the equilibrium bond length of the A state by just 0.001 Å increased the FCF from 0.016 to 0.020. Recalculation of the electronic matrix elements then yielded values of 57.2 and 58.0 cm^{-1}. These results are in good agreement with the value of 54.3 cm^{-1} calculated by Khalil et al. [4] for an internuclear separation of 2.65 Å.

Table 3
Deperturbed energies for the $X^2\Pi_{1/2}$ and $A^2\Sigma^+$ states of CaO^+.

v	$X^2\Pi_{1/2}$	$A^2\Sigma^+$
0	130.5	697.0
1	766.5	1400.6
2	1393.9	–
3	2009.1	–

Energies, given in cm^{-1}, are relative to $X^2\Pi_{3/2}$, v = 0.

Table 4
Franck-Condon Factors for the $A^2\Sigma^+$ - $X^2\Pi$ system of CaO^+.

v′/v″	0	1	2	3
0	0.212	0.292	0.231	0.139
	(0.217)	(0.295)	(0.230)	(0.137)
1	0.368	0.063	0.016	0.114
	(0.371)	(0.059)	(0.020)	(0.119)
2	0.273	0.060	0.169	0.035
	(0.270)	(0.066)	(0.169)	(0.031)
3	0.133	0.257	0.008	0.090
	(0.110)	(0.262)	(0.006)	(0.100)

The rows and columns are labeled by the vibrational quantum numbers of the $A^2\Sigma^+$ and $X^2\Pi$, respectively. The Morse potential energy curve parameters were as follows:
X; D_e = 36,491 cm^{-1}, β = 1.3773 Å^{-1}, R_e = 1.990 Å.
A; D_e = 41,207 cm^{-1}, β = 1.4257 Å^{-1}, R_e = 1.873 Å.
In each row the first series of values were calculated using equilibrium distances given above, which were taken from Ref. [4]. The values in parentheses were calculated with $R_e(A)$ increased to 1.874 Å. See text for details.

With the electronic matrix element and FCF's in hand, we can examine the initial assumption that the perturbation of the $X^2\Pi_{1/2}$, $v''=0$ level by $A^2\Sigma^+$, $v'=0$ was small enough to be neglected. In this case, the application of Eq. (2) predicts a shift of 1.3 cm^{-1}, which is slightly lower than the resolution of our spectra.

Overall, there is good agreement between the experimental data and the theoretical predictions for CaO^+. The spectrum clearly shows that the ground state is $X^2\Pi$, and the vibrational intervals for the $X^2\Pi_{3/2}$ state are remarkably close to the values predicted by Khalil et al. [4] The observed energy for the zero-point energy of $A^2\Sigma^+$ was about 30 cm^{-1} above the theoretical result. This displacement influences the $X^2\Pi_{1/2}/A^2\Sigma^+$ interaction, so the agreement between theory and experiment is not quite as good for these vibronic states. However, the error in computing the energy of the state $A^2\Sigma^+$ is impressive, given the complexity of the calculation. Similarly, it is encouraging that the spin-orbit energies and matrix elements calculated by Khalil et al. [4] are close to the observed values.

The IE value derived from the PFI-ZEKE spectrum is consistent with previous experimental determinations [1,5,6], but the error range is greatly reduced. Theoretical estimates of the IE are in acceptably good agreement, with the determination of Khalil et al. [4] being in error by just 611 cm^{-1}. Note that the measured IE for CaO is 6122 cm^{-1} above the IE for atomic Ca [19]. This is equal to the reduction of the bond energy ($D_0(CaO)$-$D_0(CaO^+)$) that accompanies ionization. As there have been previous determinations of the bond energy for neutral CaO [1–4,20], we can use this information to obtain an improved estimate for the bond energy of the cation. Irving and Dagdigian used a chemiluminescence spectrum from the reaction of $Ca(^1D)$ with O_2 to obtain a bond energy of D_0 = 4.11(7) eV for CaO [20]. Combined with the present result for the IE, this yields a D_0 value for CaO^+ of 3.35(7) eV. Previous experimental determinations of $D_0(CaO^+)$ reported values of 3.57 (5) [5], 3.30(15) [1] and 3.24(10) [1] eV, while high level theoretical calculations gave 3.29 [2], 3.33 [3] and 3.43 [4] eV. Given the uncertainties associated with both the measurements and the computational techniques, the level of agreement is reasonable.

Regarding future studies of ultra-cold CaO^+, the energies, molecular constants and FCF's reported here indicate that a reliable REMPD scheme can be constructed by exciting the $A^2\Sigma^+$ - $X^2\Pi_{3/2}$ 3–0 transition at wavelengths near 3.64 μm. The first dissociation asymptote is $Ca^+(^2S)$+$O(^3P)$, which correlates with the $X^2\Pi$ ground state and states of $^2\Sigma^-$, $^4\Sigma^-$ and $^4\Pi$ symmetry. Electric dipole transitions between $A^2\Sigma^+$ and the Σ^- states are forbidden, while the transition to the continuum region of the $X^2\Pi$ state will have a very unfavorable FCF. Consequently, the dissociation photon for the REMPD process should have sufficient energy to reach the $Ca^+(^2D)$ + $O(^3P)$ dissociation asymptote. As can be seen in Fig. 1 of Ref. [4], there are several repulsive states correlating with that limit that can be reached using fully allowed transitions. With this scheme, the wavelength of the second photon should be below 263.5 nm.

Acknowledgements

This work was supported by the Army Research Office under grant W911NF-15-1-0121.

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